CSE411 COMPUTER VISION • PROJECT TASK 3

DefectLens: Efficient ViT-Based Anomaly Detection with Multi-Criteria Token Fusion

PROJECT TEAM 7

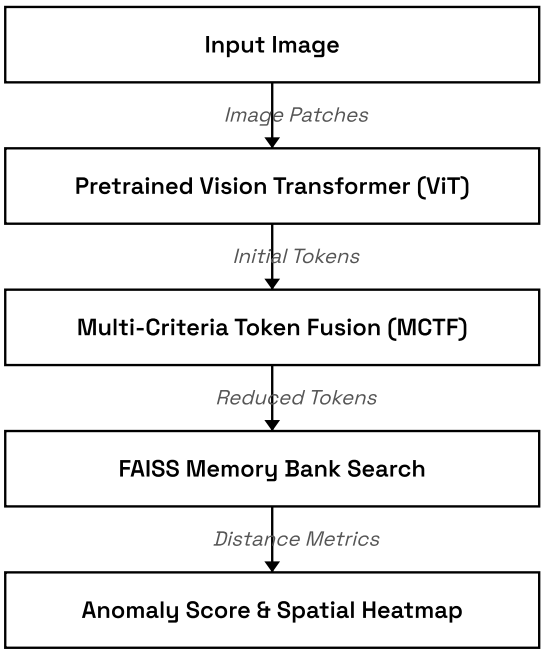
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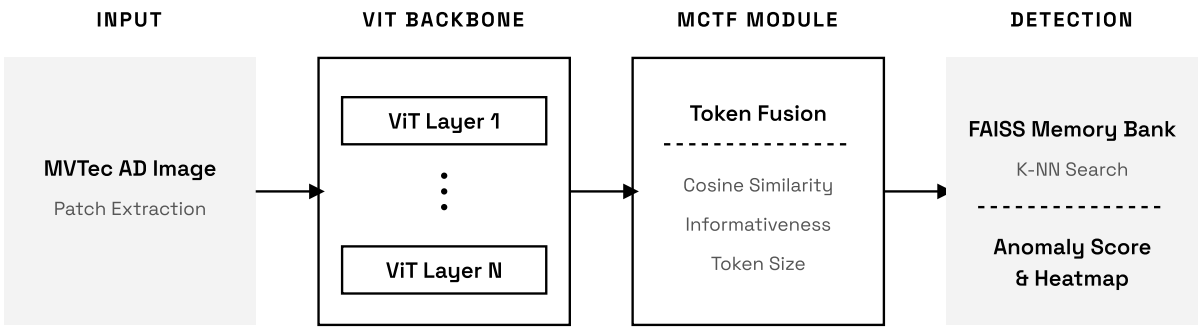
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1. System Flowchart



2. System Architecture Diagram



3. Proposed Methodology

1. Input Image Processing

High-resolution manufacturing images (e.g., from the MVTec AD dataset) are ingested into the system and standardized. They are divided into a grid of non-overlapping patches. Each patch is then linearly projected into a sequence of high-dimensional embeddings. This step serves as the foundational tokenization process, preparing the visual data for transformer-based processing.

2. Pretrained Vision Transformer (ViT) Backbone

Unlike traditional Convolutional Neural Networks (CNNs) that rely on local receptive fields, the system employs a Vision Transformer. The ViT captures robust global context immediately through self-attention mechanisms. By processing the sequence of patch embeddings across multiple layers, it establishes a rich, hierarchical feature representation capable of identifying complex, structural anomalies that standard local convolutions might miss.

3. Multi-Criteria Token Fusion (MCTF)

Standard ViTs suffer from massive computational overhead when processing high-resolution images because they compute attention over every background token equally. The MCTF module solves this by dynamically fusing semantically redundant tokens across layers. Instead of naive pooling, it uses a sophisticated bipartite soft-matching algorithm governed by three criteria: Cosine Similarity (groups alike tokens), Informativeness (protects detail regions using attention weights), and Token Size (penalizes merging large tokens). This drastically reduces total FLOPs and accelerates inference while preserving the fine-grained details necessary for detecting minuscule defects.

4. FAISS Memory Bank Nearest Neighbor Search

During the inference phase, the dimensionally reduced token features are extracted and queried against a pre-built FAISS memory bank. This memory bank serves as a repository of "normalcy," containing exclusively feature representations extracted from defect-free training data. The system utilizes an efficient K-Nearest Neighbors (KNN) algorithm to retrieve the closest matching normal features for every test patch.

5. Anomaly Score Generation & Localization

The system calculates the Euclidean distance between the extracted test features and their nearest normal neighbors to quantitatively measure deviation from the norm. These local distance metrics are aggregated to compute a comprehensive image-level anomaly score. Finally, by reverse-mapping the fused tokens back to their original spatial dimensions, the system generates a precise, pixel-level spatial heatmap. This heatmap accurately localizes defective regions for industrial review.